Modèle TK EPX

v1.2

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1 Toxicokinetics of Epoxiconazole

1.1 Results of the experiment

1.1.1 Data importation

Notations :

- `Date_start` : Start of the experiement (t0)
- `Molecule` : Substance to which the earthworm is exposed
- `Dose` : Exposure dose (mg/kg)
- `ID_recipient` : Recipient in which the soil was prepared
- `ID_camp` : ID of the earthworm campagne and generation (see XXX for more information)
- `ID` : ID of the earthworm
- `Nb_rep` : Identification number of the replicate
- `Time_point` : Time point to which the earthworm is dedicated
- `Phase` : Uptake or Elimination phase
- `Time_introduction_D0` : Time of the day when the earthworm was introduced into his contaminated cosm
- `w0` : Body weight at t0
- `w_Tp` : Body weight at the given time point
- `w_frozen` : Body weight before congelation
- `C_worm` : Measured concentration of the substance in the earthworm
- `C_soil` : Measured concentration of the substance in the soil at the given time point

1.1.2 Initial state

1.1.3 Growth curves

```
       ID            w1           Molecule               w2
 1      :  2   Min.   :0.4583   Length:124         Min.   :0.3913
 2      :  2   1st Qu.:0.5569   Class :character   1st Qu.:0.6341
 3      :  2   Median :0.6089   Mode  :character   Median :0.7323
 4      :  2   Mean   :0.6535                      Mean   :0.7362
 9      :  2   3rd Qu.:0.7134                      3rd Qu.:0.8384
 10     :  2   Max.   :0.9977                      Max.   :1.2757
```

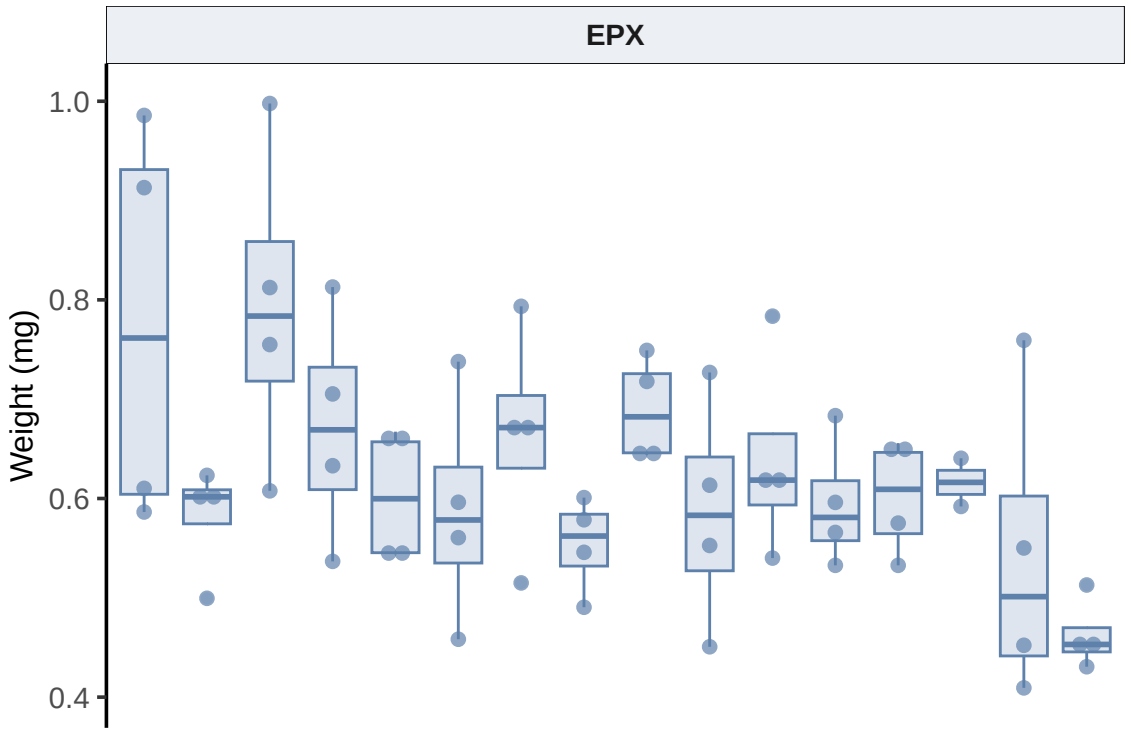


Figure 1.1: Initial weights (with gut content) of earthworms.

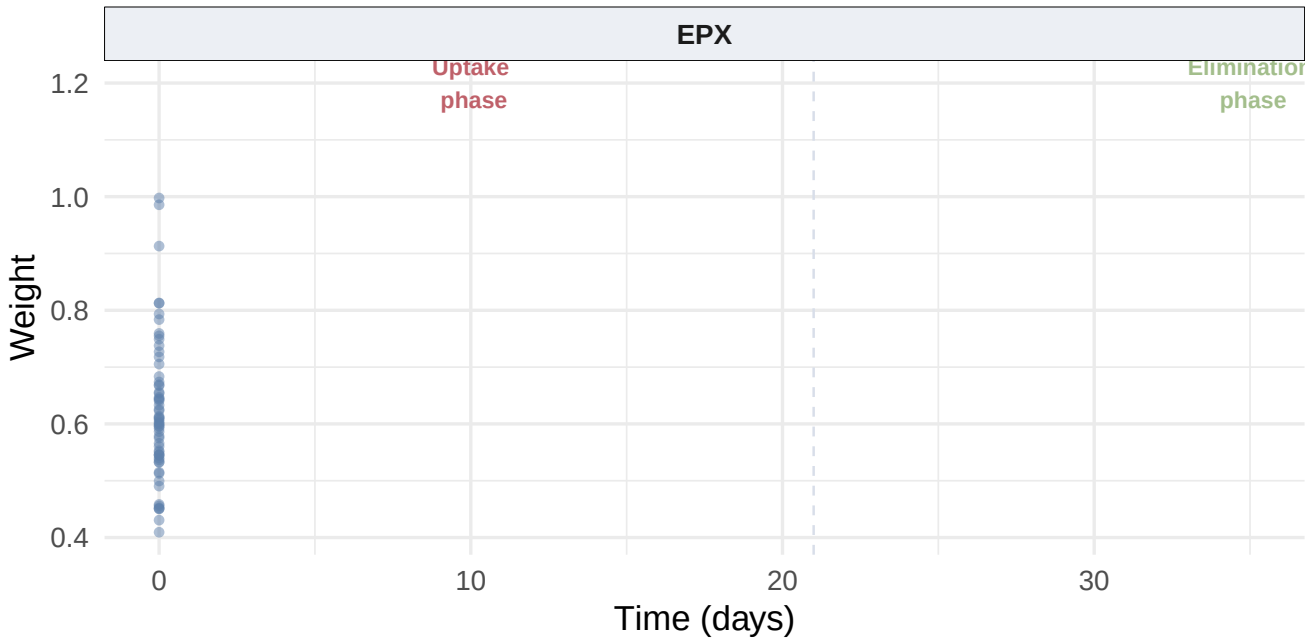


Figure 1.2: Evolution of earthworms body weight (with gut content) throughout the experiment.

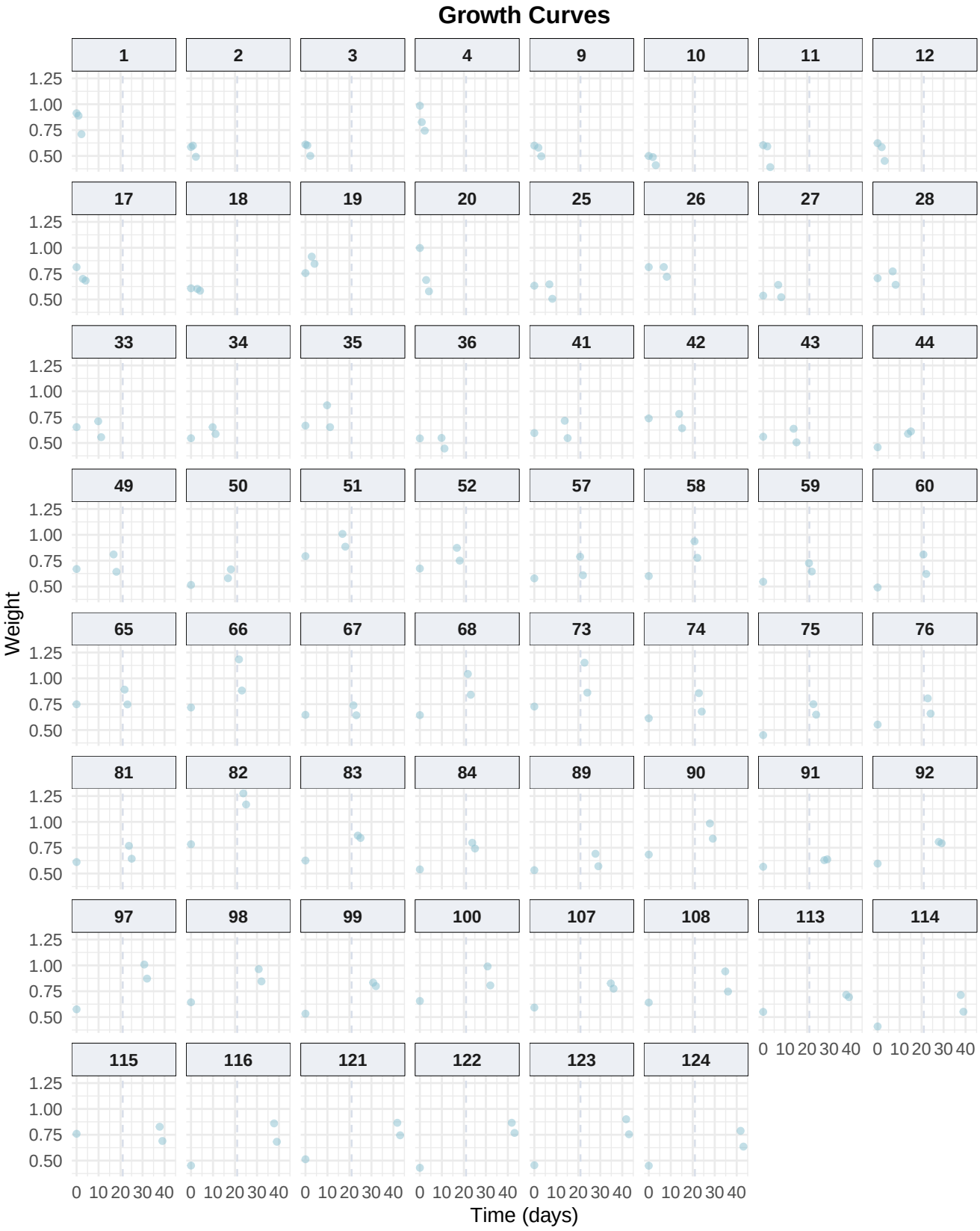


Figure 1.3: Evolution of earthworms body weight before and after fasting

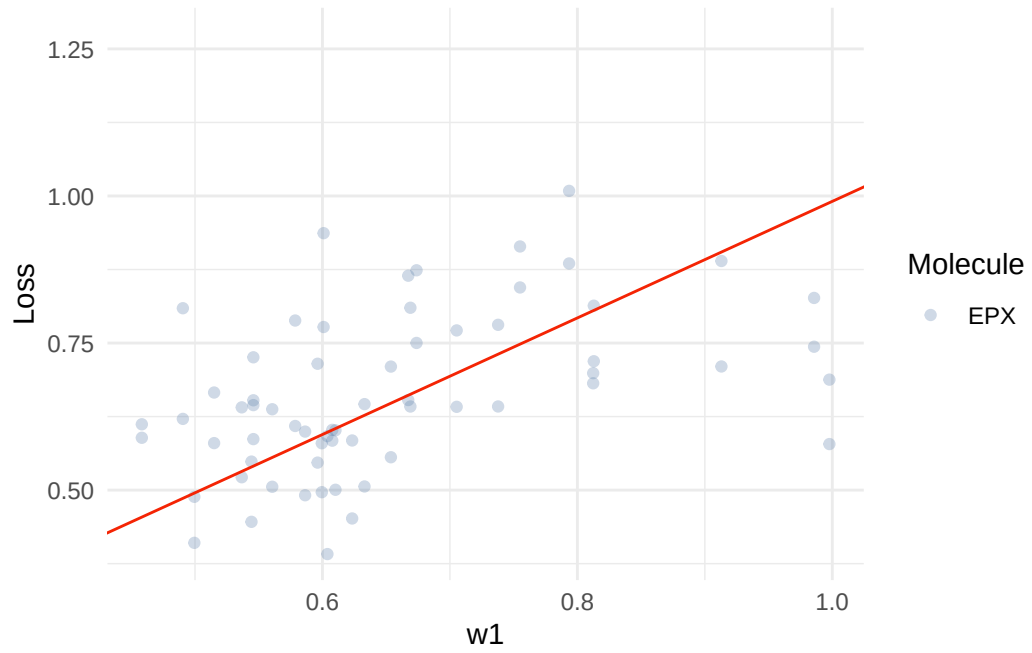
```
 (Other):112   NA's   :60
    Per_loss             Loss
 Min.   :-64.9745   Min.   :0.3913
 1st Qu.:-19.4297   1st Qu.:0.6341
 Median :  0.3992   Median :0.7323
 Mean   : -3.3081   Mean   :0.7362
 3rd Qu.: 13.2139   3rd Qu.:0.8384
 Max.   : 42.0367   Max.   :1.2757
 NA's   :60


Call:
lm(formula = Loss ~ w1 + 0, data = df_TK_fasting)

Residuals:
     Min       1Q   Median       3Q      Max
-0.41029 -0.08700  0.00413  0.10965  0.34139

Coefficients:
   Estimate Std. Error t value Pr(>|t|)
w1  0.99087    0.02659   37.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.142 on 63 degrees of freedom
  (60 observations deleted due to missingness)
Multiple R-squared:  0.9566,    Adjusted R-squared:  0.9559
F-statistic:  1388 on 1 and 63 DF,  p-value: < 2.2e-16
```



!!!!!!!! Growth rates !!!!!

1.1.4 Measured internal concentrations

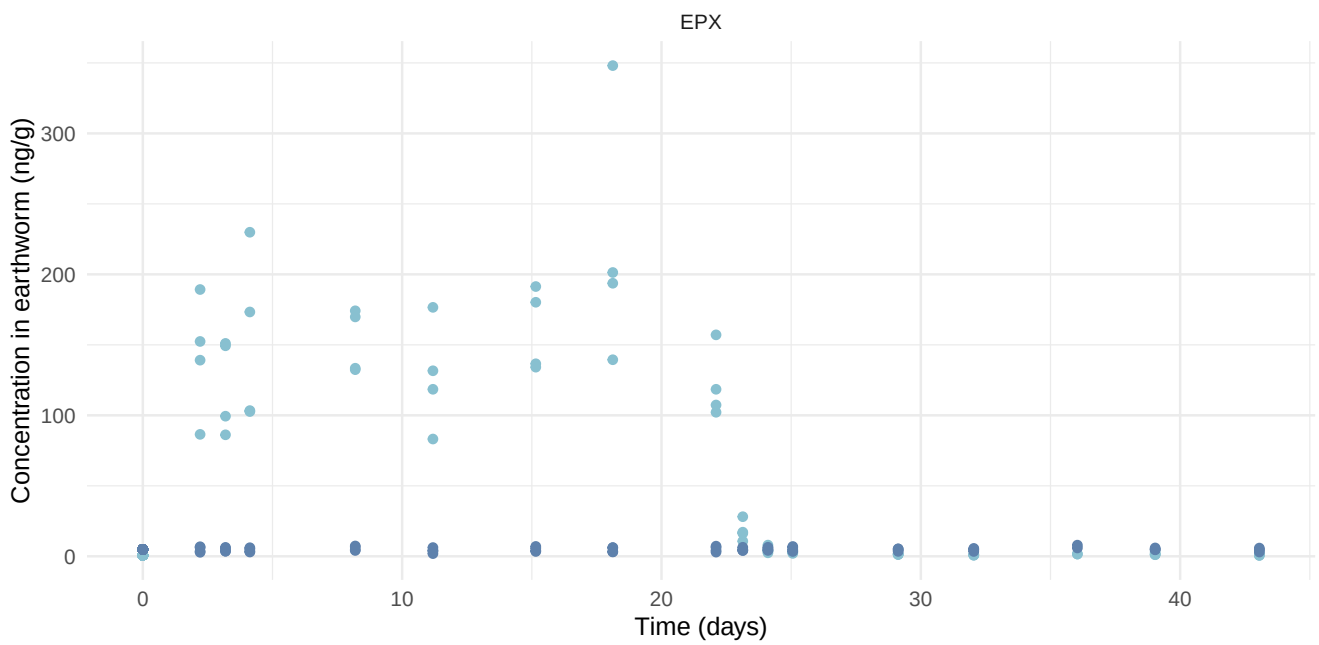


Figure 1.4: Measurements of EPX concentration in earthworms exposed to 10 mg/kg of EPX and related weight measurement of earthworms. Points represent the experimental data, the blue curve and the transparent blue band are produced with geom_smooth.

1.2 Modeling of the toxicokinetics

The first step is to retrieve the information of the MCMC simulation experiments.

The inputs :

- Input files must a name like "xxxxx.1.in"
- Output files must a name like "xxxxx.1.out"

We can retrieve different information from the model :

- The number of chains used : 3
- The number of parameters : 6
- The number of experiments : 62
- The number of last iterations used to describe the posterior distribution : 10^4
- The number of last iterations used to create the `tab_setpoint.out` file : 333

Warning : Run the command `./mcsim.m_TK_EPX TK_EPX_Setpoint.in` in the terminal while being in the folder `mod/TK_EPX` to create the `Setpoints.out` now. Do the same with `./mcsim.m_TK_EPX TK_EPX_Setpoint_ind.in` to create the `Setpoints_ind.out` and with `./mcsim.m_TK_EPX TK_EPX_Setpoint_ind_fu` to create the `Setpoints_ind_full.out`.

1.2.1 Priors used for the model

```
# A tibble: 6 x 6
  Nom         Distribution     P1          P2      P3    P4
  <chr>       <chr>         <dbl>       <dbl>   <dbl> <dbl>
1 kuEPX       LogUniform   0.0001 100000      NA         NA
2 keEPX       LogUniform   0.0001 100000      NA         NA
3 a_growth    TruncNormal  0.005       0.005  -0.5        1
4 Vr_a_growth TruncNormal  0.0015      0.0015  0          1
5 Sigma_W     TruncNormal  0.03        0.01    0.0001     1
6 Sigma_CiEPX TruncNormal  3           3       1       1000
```

1.2.2 Chains

We can also take a look at the likelihood and the deviance ratio.

Correlations :

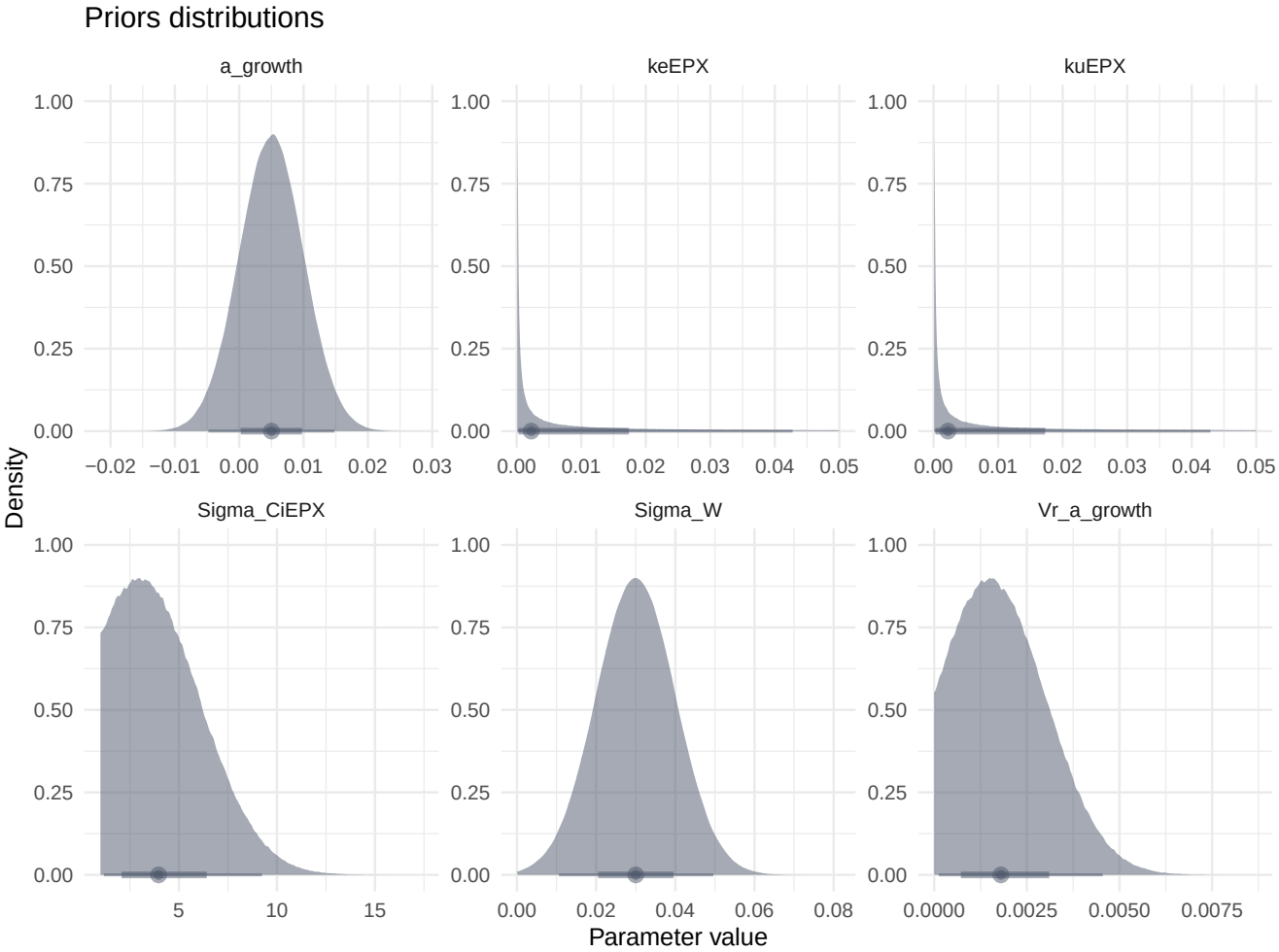


Figure 1.5: Distributions of the priors.

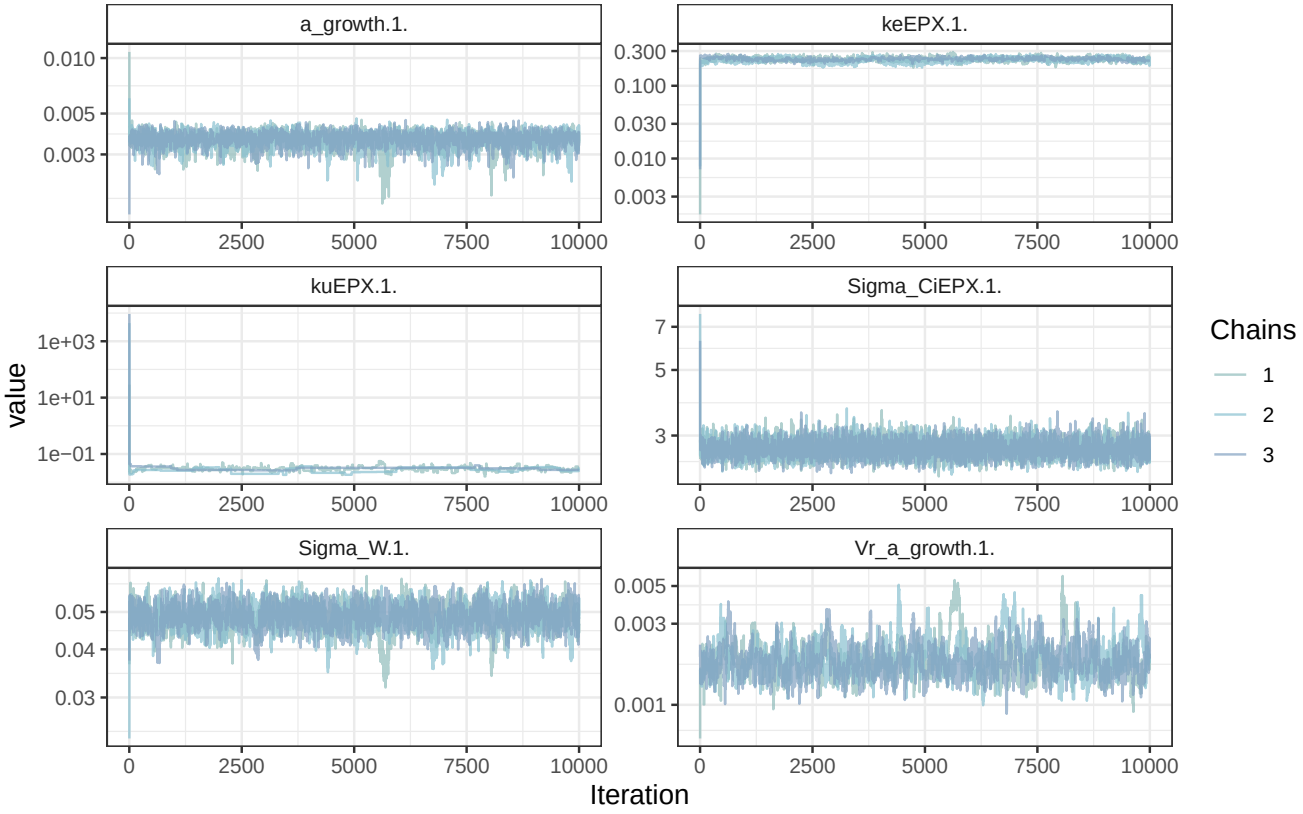


Figure 1.6: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

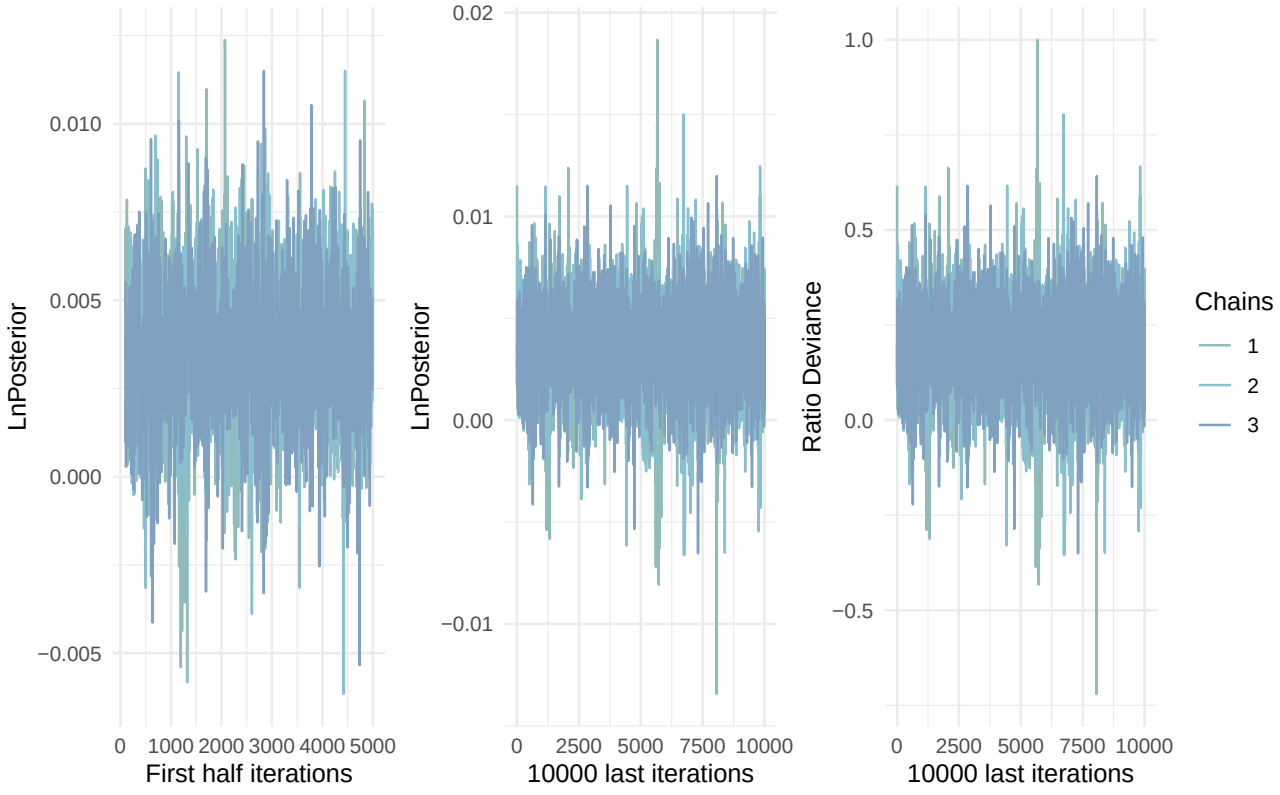


Figure 1.7: Evolution of likelihood and deviance ratio through the iterations.

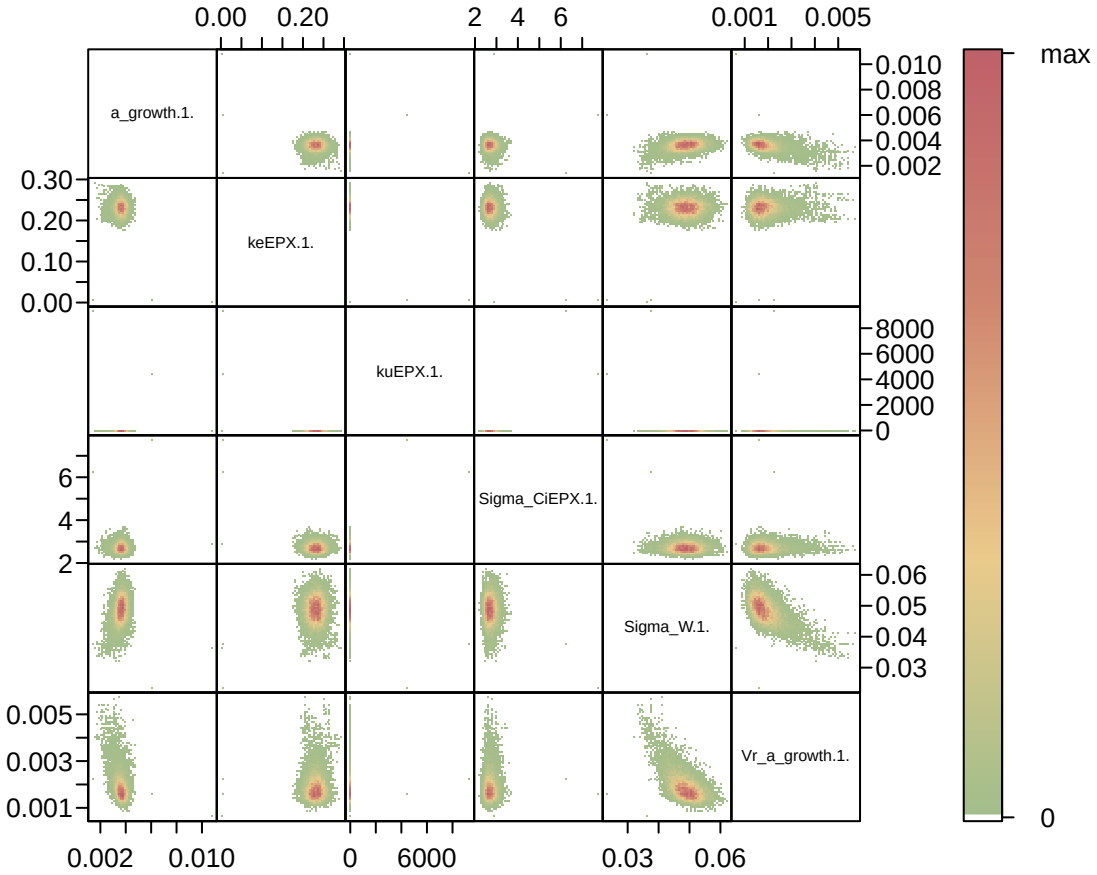


Figure 1.8: Parameter correlations.

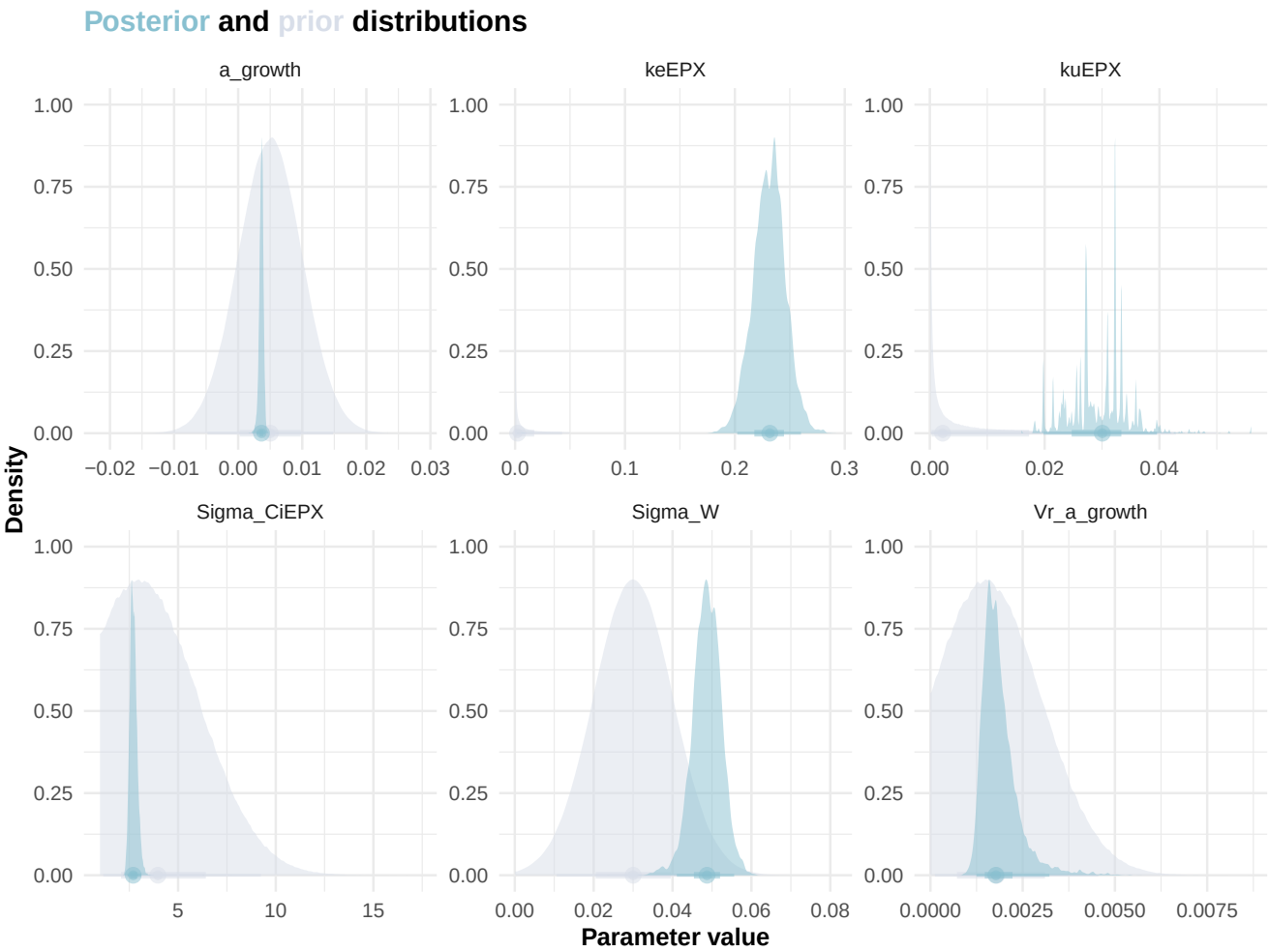


Figure 1.9: Distributions of the priors and the posteriors.

1.2.3 Posteriors and parameter estimation results

```
# A tibble: 7 x 7
  kuEPX.1. keEPX.1. a_growth.1. Vr_a_growth.1. Sigma_W.1. Sigma_CiEPX.1.    BCF
     <dbl>    <dbl>       <dbl>          <dbl>      <dbl>          <dbl>  <dbl>
1    0.032   0.24       0.0039         0.0012      0.048            2.8   0.12
2    0.02    0.2        0.0029         0.0012      0.041            2.4   0.094
3    0.04    0.26       0.0042         0.0032      0.056            3.1   0.16
4    0.016   0.0017     0.0014         0.00063     0.023            2.2   0.076
5 9300       0.29       0.011          0.0057      0.062            7.7  17
6    0.49    0.23       0.0036         0.0019      0.049            2.7   0.13
7   59       0.015      0.00033        0.00051     0.0037           0.18  0.11
```

1.2.4 Comparison of observed vs. predicted data

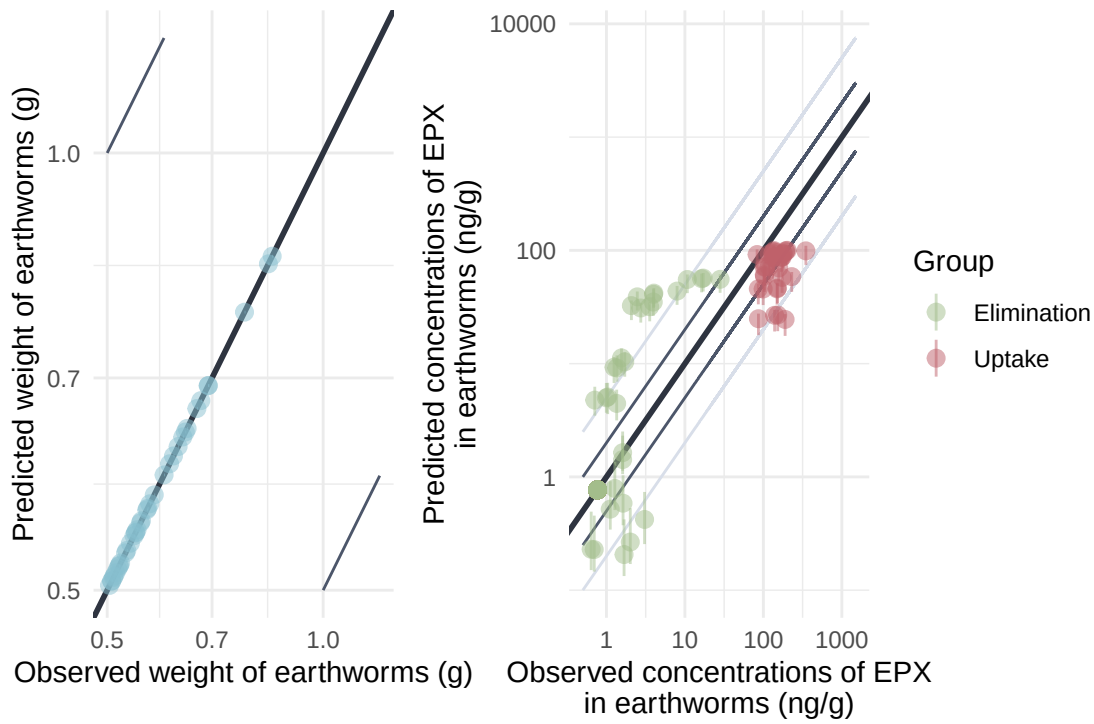


Figure 1.10: Results of the model fit on experimental data. Predicted values are represented with points with their respective credibility interval of 95%. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.

```
For weights :

 Percentage of prediction in fold 2 =  100 %
 Percentage of prediction in fold 5 =  100 %
```

```
For internal concentrations :

 Percentage of prediction in fold 2 =  69.4 %
 Percentage of prediction in fold 5 =  83.1 %
```

1.2.5 Full individual trajectories

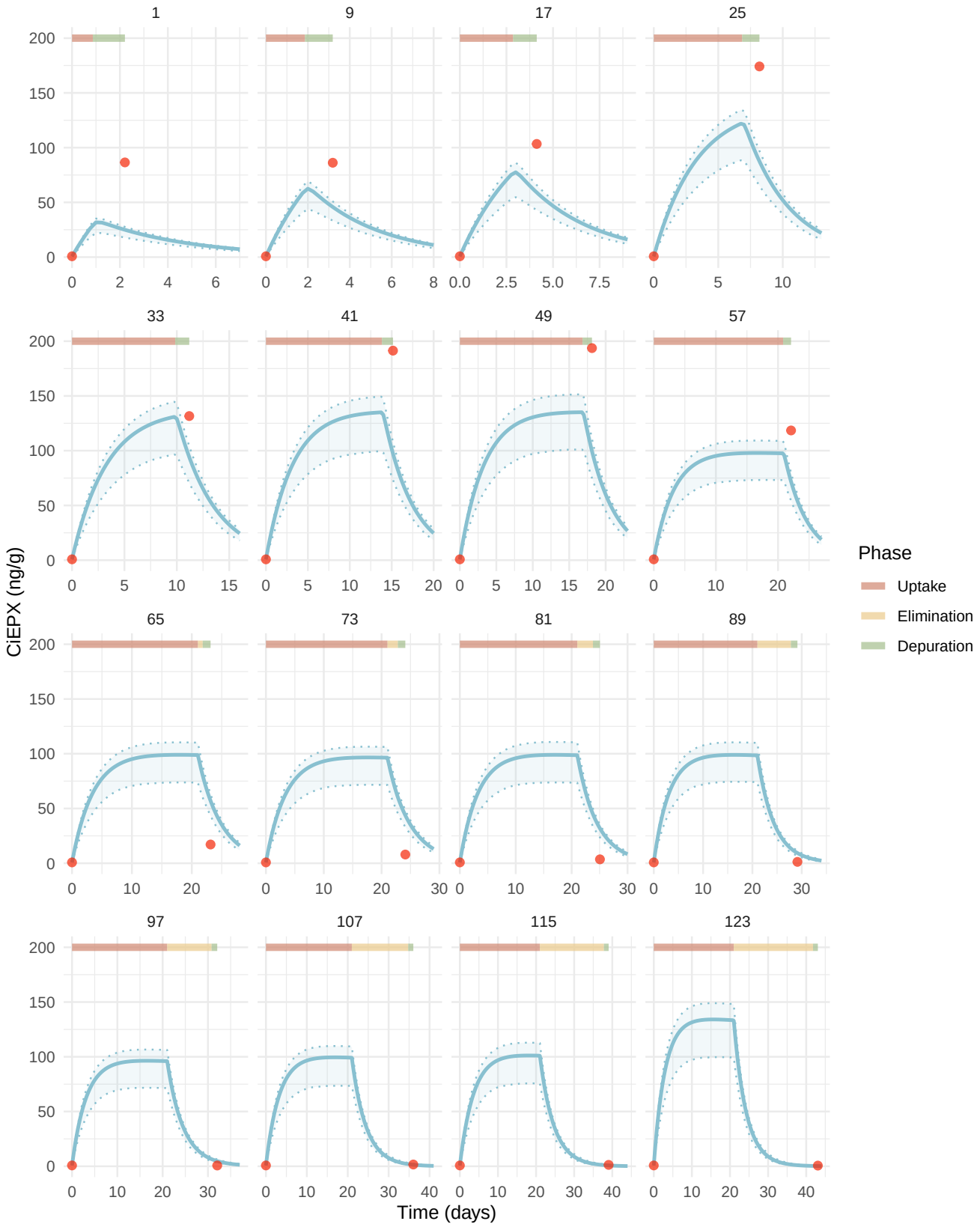


Figure 1.11: Estimated toxicokinetic of the EPX concentration in earthworms exposed to 1000 ng/g of EPX. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

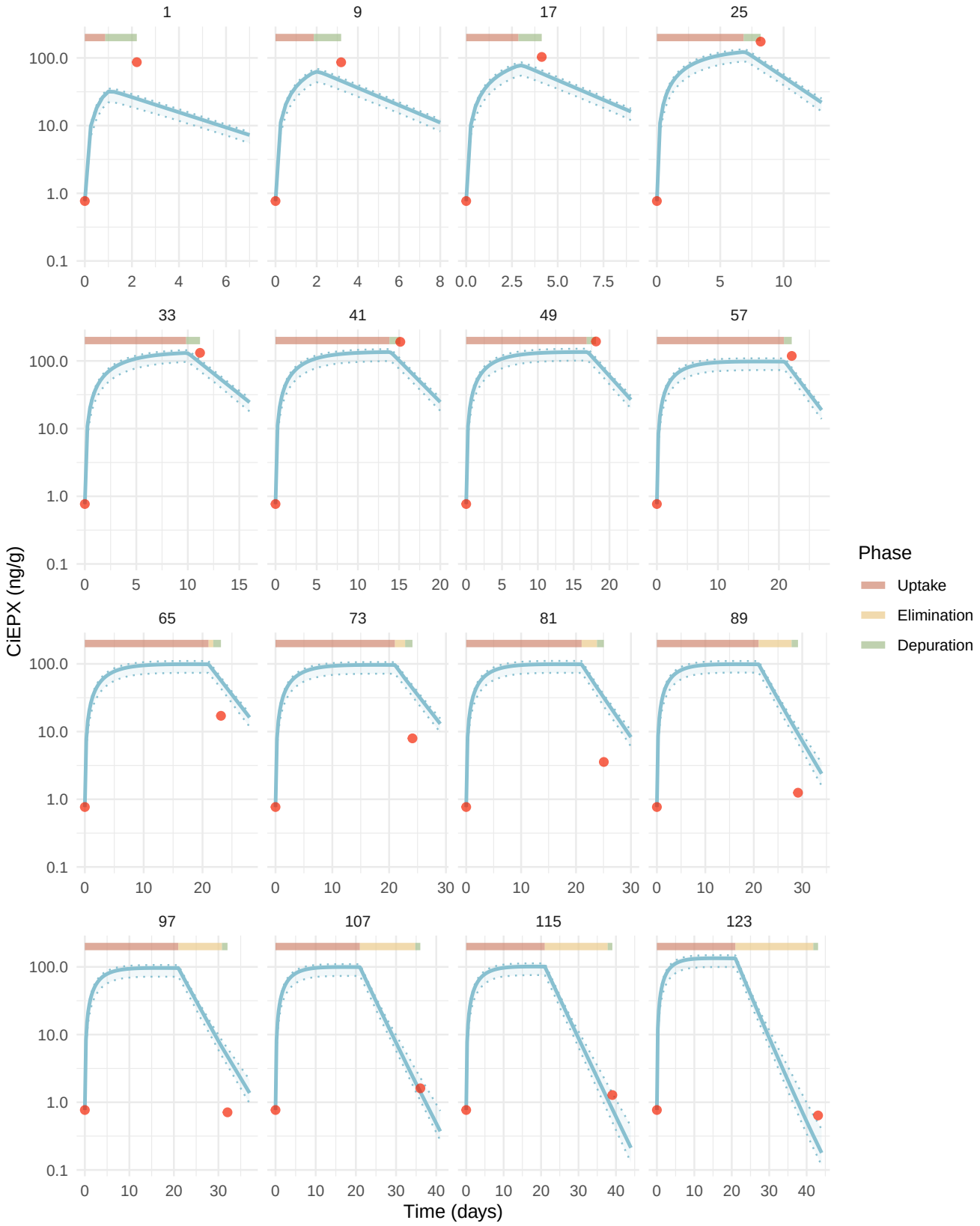


Figure 1.12: Estimated toxicokinetic of the EPX concentration in earthworms exposed to 1000 ng/g of EPX. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

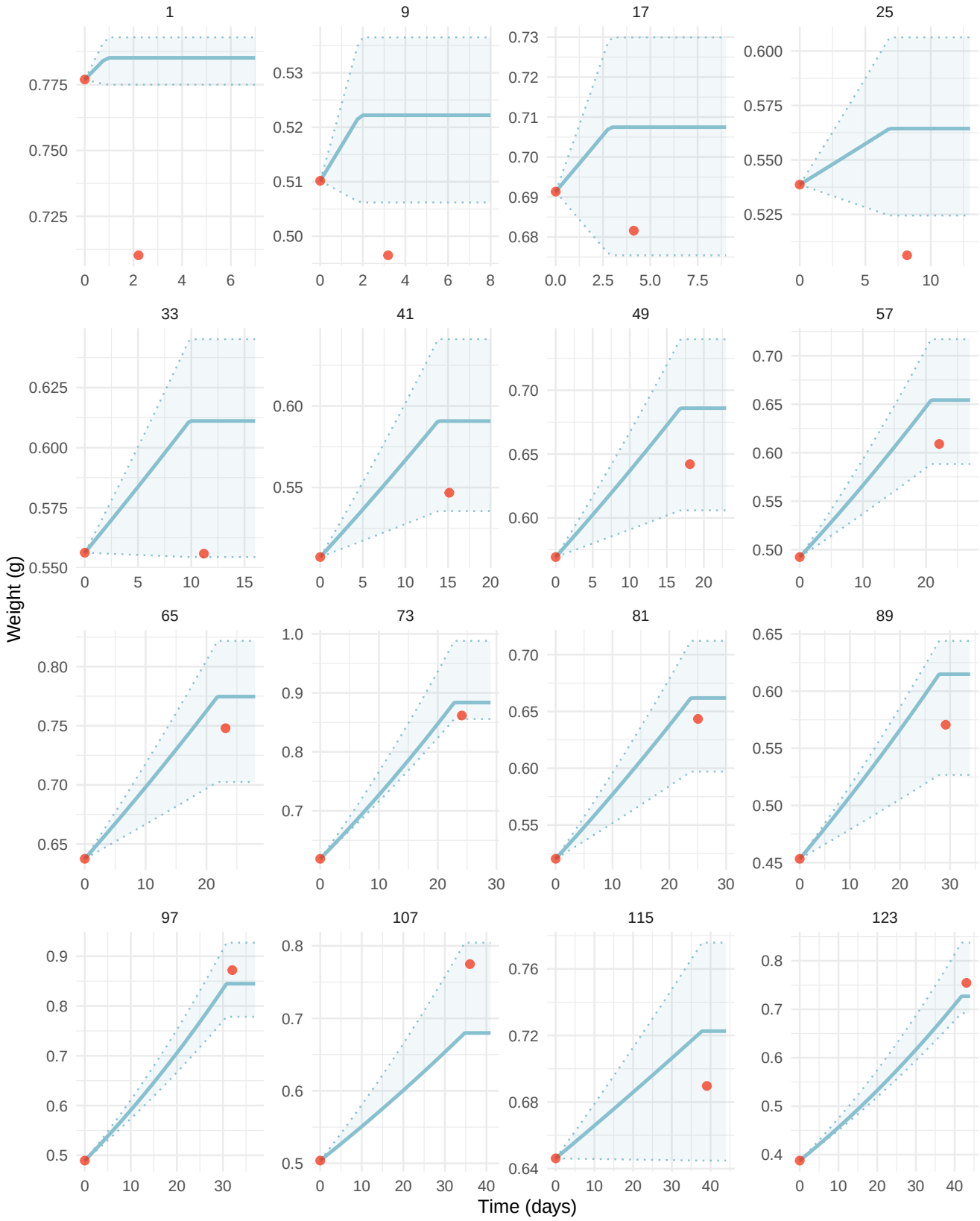


Figure 1.13: Weight predictions

1.2.6 Simulation of the toxicodynamic in earthworms throughout the exposition

The goal here is to recreate the concentration dynamic in the average earthworm during our exposition scheme but without the fasting phase that was required for chemical analysis and during which elimination was happening.

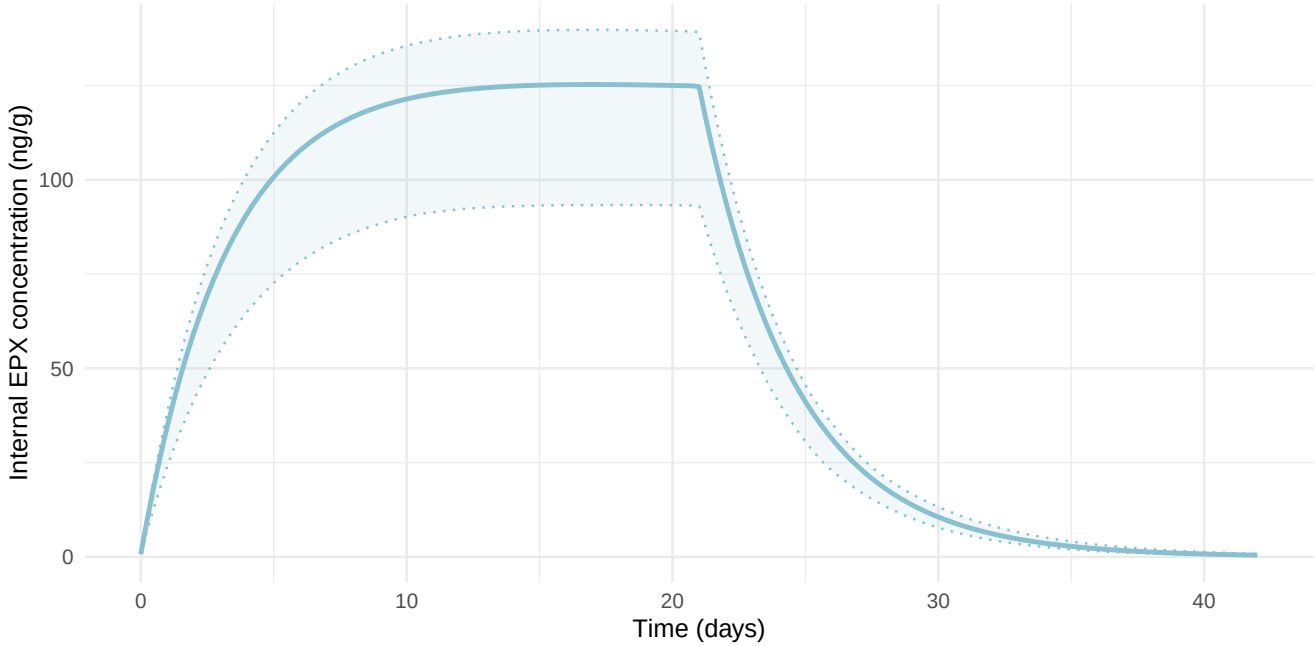


Figure 1.14: Estimated toxicokinetic of the EPX concentration in earthworms exposed to 0.1 mg/kg of EPX. Blue points represent the experimental data, the blue curve represents the estimated toxicokinetic curve and the transparent blue band corresponds to the credibility interval of the model.